Validacao de Estado

Demonstração (via redis-cli) de que o banco contém informações atualizadas e sem duplicidade por entregador

Evidência 1

No meu projeto tenho 4 entregadores, então:

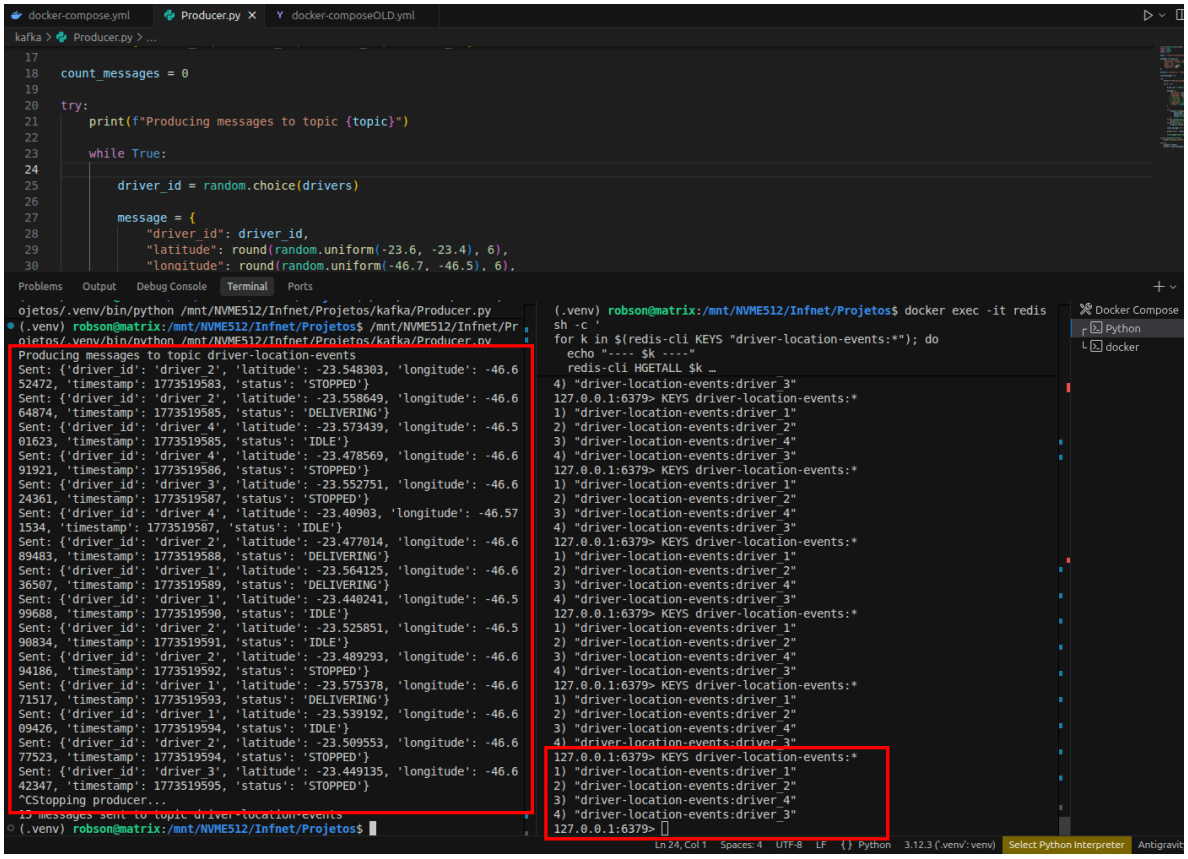
```
robson@matrix:/mnt/NVME512/Infnet/Projetos$ docker exec -it redis re
dis-cli
127.0.0.1:6379> KEYS driver-location-events:*
1) "driver-location-events:driver_2"
2) "driver-location-events:driver_3"
3) "driver-location-events:driver_1"
4) "driver-location-events:driver_4"
127.0.0.1:6379>
Ln 199, Col 21 (73 selected)  Spaces: 2  UTF-8  LF  {} Compose  Antigrav
```

Neste exemplo a última posição do driver 1

```
127.0.0.1:6379> HGETALL driver-location-events:driver_1
 1) "driver_id"
 2) "driver_1"
 3) "latitude"
 4) "-23.471841"
 5) "longitude"
 6) "-46.667937"
 7) "timestamp"
 8) "1773288861"
 9) "status"
10) "STOPPED"
127.0.0.1:6379>
Ln 199, Col 21 (73 selected)  Spaces: 2  UTF-8  LF  {} Compose  Antigravity - Settings
```

Evidência 2

Rodando o Produtor e verificando no Redis



Evidência 3 - Detalhe dos dados

```
Problems   Output   Debug Console   Terminal   Ports

(.venv) robson@matrix:/mnt/NVME512/Infnet/Projetos$ /mnt/NVME512/Infnet/Pr
ojetos/.venv/bin/python /mnt/NVME512/Infnet/Projetos/kafka/Producer.py
45284, 'timestamp': 1773519533, 'status': 'DELIVERING'}
Sent: {'driver_id': 'driver_3', 'latitude': -23.438255, 'longitude': -46.6
84759, 'timestamp': 1773519534, 'status': 'DELIVERING'}
Sent: {'driver_id': 'driver_2', 'latitude': -23.48899, 'longitude': -46.64
9325, 'timestamp': 1773519535, 'status': 'STOPPED'}
Sent: {'driver_id': 'driver_4', 'latitude': -23.464547, 'longitude': -46.5
93231, 'timestamp': 1773519535, 'status': 'DELIVERING'}
Sent: {'driver_id': 'driver_1', 'latitude': -23.506359, 'longitude': -46.5
81499, 'timestamp': 1773519535, 'status': 'DELIVERING'}
Sent: {'driver_id': 'driver_3', 'latitude': -23.485122, 'longitude': -46.6
04261, 'timestamp': 1773519536, 'status': 'IDLE'}
^CStopping producer...
463 messages sent to topic driver-location-events
(.venv) robson@matrix:/mnt/NVME512/Infnet/Projetos$ /mnt/NVME512/Infnet/Pr
ojetos/.venv/bin/python /mnt/NVME512/Infnet/Projetos/kafka/Producer.py
Producing messages to topic driver-location-events
Sent: {'driver_id': 'driver_2', 'latitude': -23.548303, 'longitude': -46.6
52472, 'timestamp': 1773519583, 'status': 'STOPPED'}
Sent: {'driver_id': 'driver_2', 'latitude': -23.558649, 'longitude': -46.6
64874, 'timestamp': 1773519585, 'status': 'DELIVERING'}
Sent: {'driver_id': 'driver_4', 'latitude': -23.573439, 'longitude': -46.5
01623, 'timestamp': 1773519585, 'status': 'IDLE'}
Sent: {'driver_id': 'driver_4', 'latitude': -23.478569, 'longitude': -46.6
91921, 'timestamp': 1773519586, 'status': 'STOPPED'}
Sent: {'driver_id': 'driver_3', 'latitude': -23.552751, 'longitude': -46.6
24361, 'timestamp': 1773519587, 'status': 'STOPPED'}
Sent: {'driver_id': 'driver_4', 'latitude': -23.40903, 'longitude': -46.57
1534, 'timestamp': 1773519587, 'status': 'IDLE'}
Sent: {'driver_id': 'driver_2', 'latitude': -23.477014, 'longitude': -46.6
89483, 'timestamp': 1773519588, 'status': 'DELIVERING'}
Sent: {'driver_id': 'driver_1', 'latitude': -23.564125, 'longitude': -46.6
36507, 'timestamp': 1773519589, 'status': 'DELIVERING'}
Sent: {'driver_id': 'driver_1', 'latitude': -23.440241, 'longitude': -46.5
99688, 'timestamp': 1773519590, 'status': 'IDLE'}
Sent: {'driver_id': 'driver_2', 'latitude': -23.525851, 'longitude': -46.5
90834, 'timestamp': 1773519591, 'status': 'IDLE'}
Sent: {'driver_id': 'driver_2', 'latitude': -23.489293, 'longitude': -46.6
94186, 'timestamp': 1773519592, 'status': 'STOPPED'}
Sent: {'driver_id': 'driver_1', 'latitude': -23.575378, 'longitude': -46.6
71517, 'timestamp': 1773519593, 'status': 'DELIVERING'}
Sent: {'driver_id': 'driver_1', 'latitude': -23.539192, 'longitude': -46.6
09426, 'timestamp': 1773519594, 'status': 'IDLE'}
Sent: {'driver_id': 'driver_2', 'latitude': -23.509553, 'longitude': -46.6
77523, 'timestamp': 1773519594, 'status': 'STOPPED'}
Sent: {'driver_id': 'driver_3', 'latitude': -23.449135, 'longitude': -46.6
42347, 'timestamp': 1773519595, 'status': 'STOPPED'}
^CStopping producer...
15 messages sent to topic driver-location-events
(.venv) robson@matrix:/mnt/NVME512/Infnet/Projetos$

(.venv) robson@matrix:/mnt/NVME512/Infnet/Projetos$ docker exec -it redis
sh -c '
for k in $(redis-cli KEYS "driver-location-events:*"); do
  echo "---- $k ----"
  redis-cli HGETALL $k …
'
---- driver-location-events:driver_1 ----
 1) "driver_id"
 2) "driver_1"
 3) "latitude"
 4) "-23.539192"
 5) "longitude"
 6) "-46.609426"
 7) "timestamp"
 8) "1773519594"
 9) "status"
10) "IDLE"
---- driver-location-events:driver_2 ----
 1) "driver_id"
 2) "driver_2"
 3) "latitude"
 4) "-23.509553"
 5) "longitude"
 6) "-46.677523"
 7) "timestamp"
 8) "1773519594"
 9) "status"
10) "STOPPED"
---- driver-location-events:driver_4 ----
 1) "driver_id"
 2) "driver_4"
 3) "latitude"
 4) "-23.40903"
 5) "longitude"
 6) "-46.571534"
 7) "timestamp"
 8) "1773519587"
 9) "status"
10) "IDLE"
---- driver-location-events:driver_3 ----
 1) "driver_id"
 2) "driver_3"
 3) "latitude"
 4) "-23.449135"
 5) "longitude"
 6) "-46.642347"
 7) "timestamp"
 8) "1773519595"
 9) "status"
10) "STOPPED"
(.venv) robson@matrix:/mnt/NVME512/Infnet/Projetos$
```

"latitude", "longitude", "timestamp", "status" com mesmo valor final

Evidência 4

```
(.venv) robson@matrix:/mnt/NVME512/Infnet/Projetos$ /mnt/NVME512/Infnet/Pr
ojetos/.venv/bin/python /mnt/NVME512/Infnet/Projetos/kafka/Producer.py
79457, 'timestamp': 1773520217, 'status': 'DELIVERING'}
Sent: {'driver_id': 'driver_1', 'latitude': -23.493755, 'longitude': -46.6
46228, 'timestamp': 1773520218, 'status': 'STOPPED'}
Sent: {'driver_id': 'driver_1', 'latitude': -23.493523, 'longitude': -46.6
60349, 'timestamp': 1773520219, 'status': 'IDLE'}
Sent: {'driver_id': 'driver_4', 'latitude': -23.436573, 'longitude': -46.6
92145, 'timestamp': 1773520220, 'status': 'IDLE'}
Sent: {'driver_id': 'driver_2', 'latitude': -23.461303, 'longitude': -46.5
16511, 'timestamp': 1773520220, 'status': 'STOPPED'}
Sent: {'driver_id': 'driver_4', 'latitude': -23.459253, 'longitude': -46.5
76035, 'timestamp': 1773520221, 'status': 'DELIVERING'}
Sent: {'driver_id': 'driver_2', 'latitude': -23.45671, 'longitude': -46.60
0687, 'timestamp': 1773520222, 'status': 'DELIVERING'}
Sent: {'driver_id': 'driver_1', 'latitude': -23.487872, 'longitude': -46.6
56967, 'timestamp': 1773520223, 'status': 'DELIVERING'}
Sent: {'driver_id': 'driver_2', 'latitude': -23.587711, 'longitude': -46.5
58873, 'timestamp': 1773520223, 'status': 'STOPPED'}
Sent: {'driver_id': 'driver_4', 'latitude': -23.550791, 'longitude': -46.6
9617, 'timestamp': 1773520224, 'status': 'DELIVERING'}
Sent: {'driver_id': 'driver_2', 'latitude': -23.425477, 'longitude': -46.6
00725, 'timestamp': 1773520225, 'status': 'DELIVERING'}
Sent: {'driver_id': 'driver_2', 'latitude': -23.540951, 'longitude': -46.6
94173, 'timestamp': 1773520225, 'status': 'STOPPED'}
Sent: {'driver_id': 'driver_4', 'latitude': -23.581764, 'longitude': -46.6
23658, 'timestamp': 1773520226, 'status': 'IDLE'}
Sent: {'driver_id': 'driver_2', 'latitude': -23.4141, 'longitude': -46.586
827, 'timestamp': 1773520227, 'status': 'IDLE'}
Sent: {'driver_id': 'driver_2', 'latitude': -23.553021, 'longitude': -46.6
22207, 'timestamp': 1773520227, 'status': 'DELIVERING'}
Sent: {'driver_id': 'driver_4', 'latitude': -23.46542, 'longitude': -46.55
9476, 'timestamp': 1773520228, 'status': 'STOPPED'}
Sent: {'driver_id': 'driver_2', 'latitude': -23.465245, 'longitude': -46.5
4434, 'timestamp': 1773520229, 'status': 'STOPPED'}
Sent: {'driver_id': 'driver_1', 'latitude': -23.431744, 'longitude': -46.6
37806, 'timestamp': 1773520230, 'status': 'DELIVERING'}
^CStopping producer...
61 messages sent to topic driver-location-events
(.venv) robson@matrix:/mnt/NVME512/Infnet/Projetos$

(.venv) robson@matrix:/mnt/NVME512/Infnet/Projetos$ docker exec -it redis
redis-cli
iver_1" "driver_id" "driver_1" "latitude" "-23.431744" "longitude" "-46.63
7806" "timestamp" "1773520230" "status" "DELIVERING"
1773520230.606719 [0 172.20.0.120:56162] "MSET" "com.redis.kafka.connect.s
ink.offset.driver-location-events.0" "{\"topic\":\"driver-location-events\
",\"partition\":0,\"offset\":10077}"
^C(190.15s)
127.0.0.1:6379> monitor
OK
^C(2.98s)
127.0.0.1:6379>
(.venv) robson@matrix:/mnt/NVME512/Infnet/Projetos$ HGETALL driver-locatio
n-events:driver_3
HGETALL: command not found
(.venv) robson@matrix:/mnt/NVME512/Infnet/Projetos$ docker exec -it redis
redis-cli
127.0.0.1:6379> HGETALL driver-location-events:driver_3
 1) "driver_id"
 2) "driver_3"
 3) "latitude"
 4) "-23.498451"
 5) "longitude"
 6) "-46.54057"
 7) "timestamp"
 8) "1773520216"
 9) "status"
10) "DELIVERING"
127.0.0.1:6379> HGETALL driver-location-events:driver_2
 1) "driver_id"
 2) "driver_2"
 3) "latitude"
 4) "-23.465245"
 5) "longitude"
 6) "-46.54434"
 7) "timestamp"
 8) "1773520229"
 9) "status"
10) "STOPPED"
127.0.0.1:6379>
```

Comandos usados:

```cmd
docker exec -it redis sh -c '
for k in $(redis-cli KEYS "driver-location-events:*"); do
  echo "---- $k ----"
  redis-cli HGETALL $k
done
'
```

```cmd
>KEYS driver-location-events:*
```

```cmd
>HGETALL driver-location-events:driver_2
```